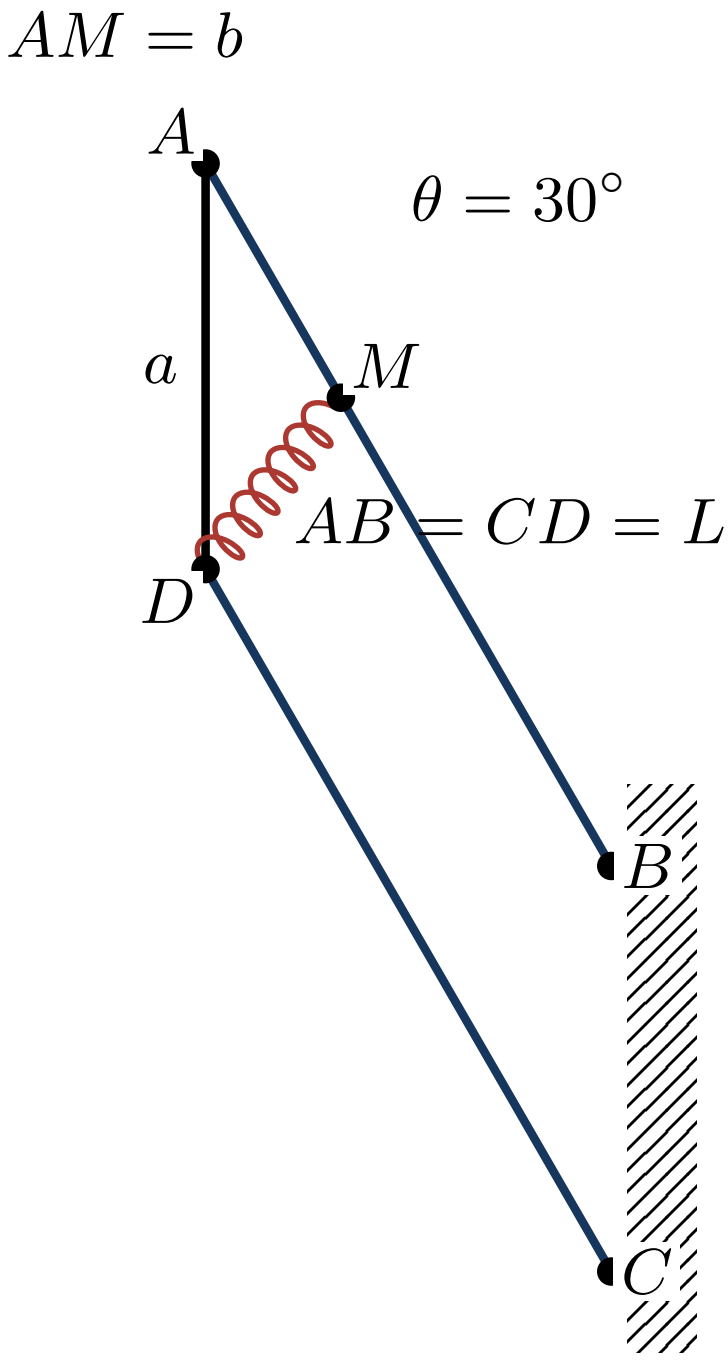
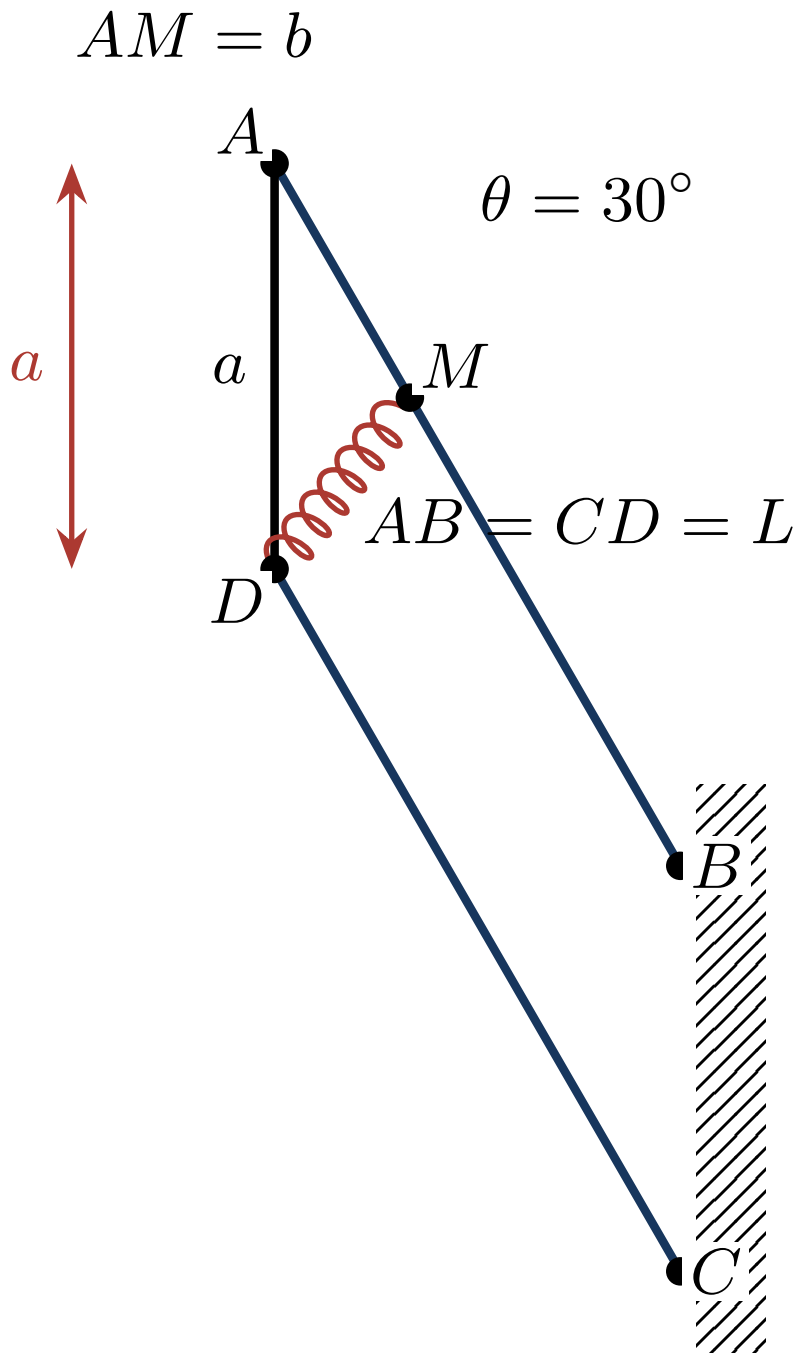


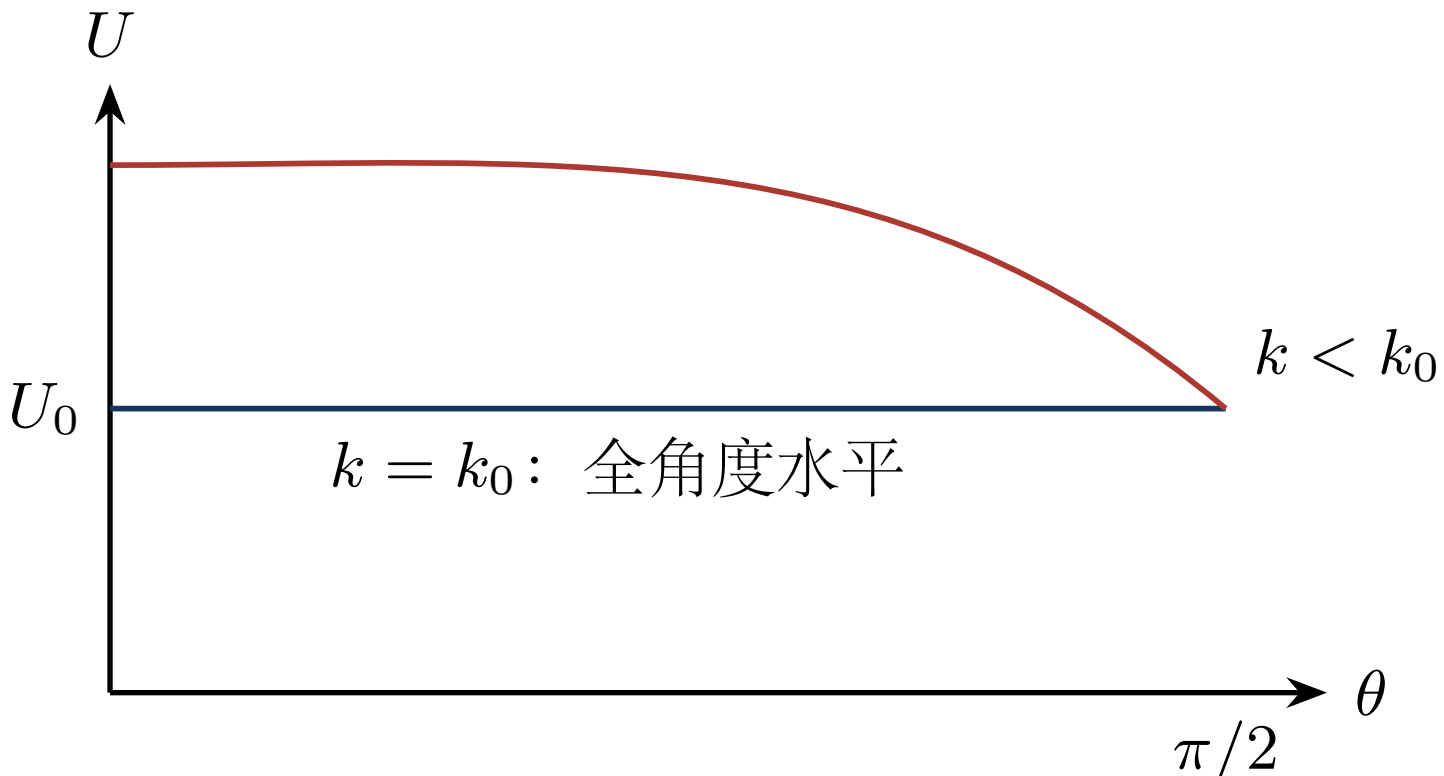
两横杆各长 L；墙上铰距 a 未给数值

弹簧自然长为零，连接 M, D

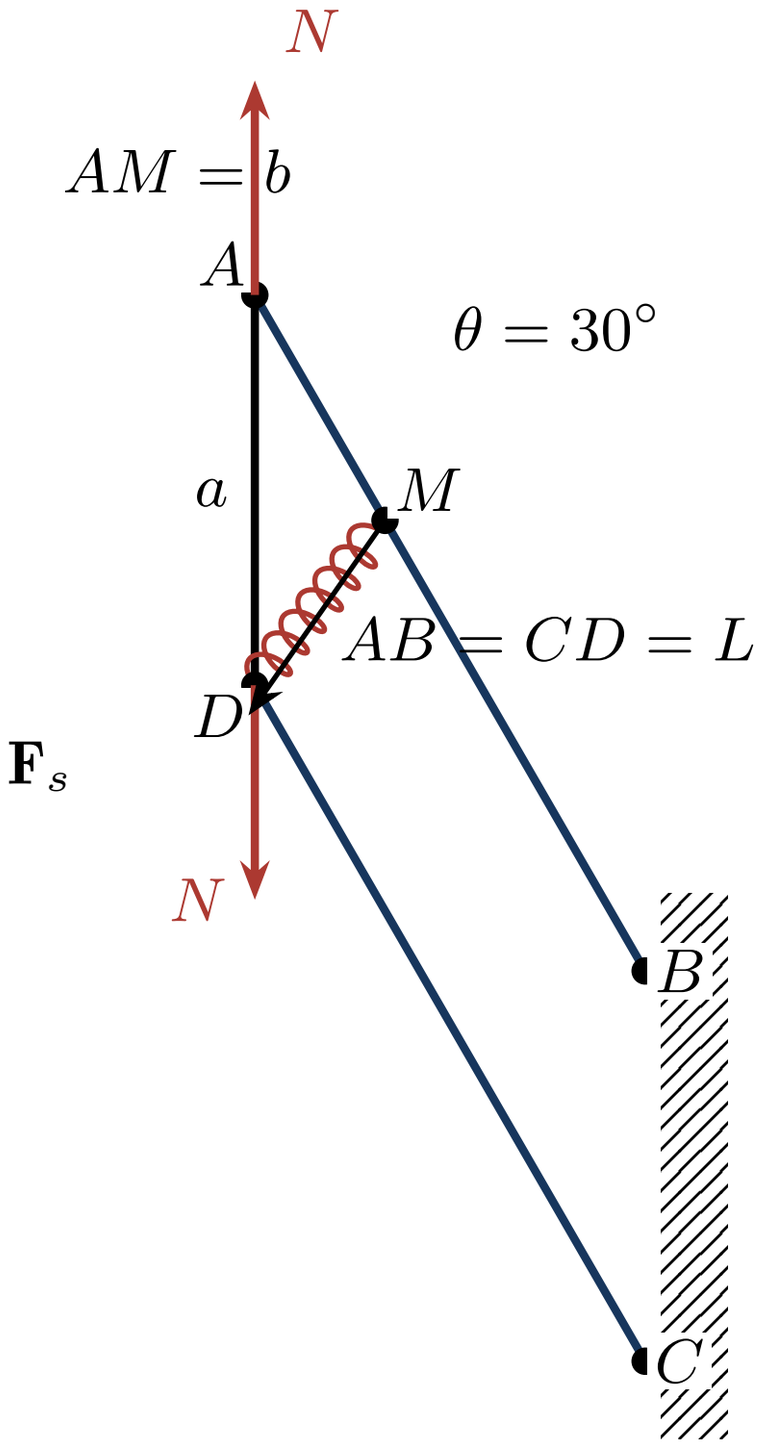


$$MD^2 = a^2 + b^2 - 2ab\cos\theta$$

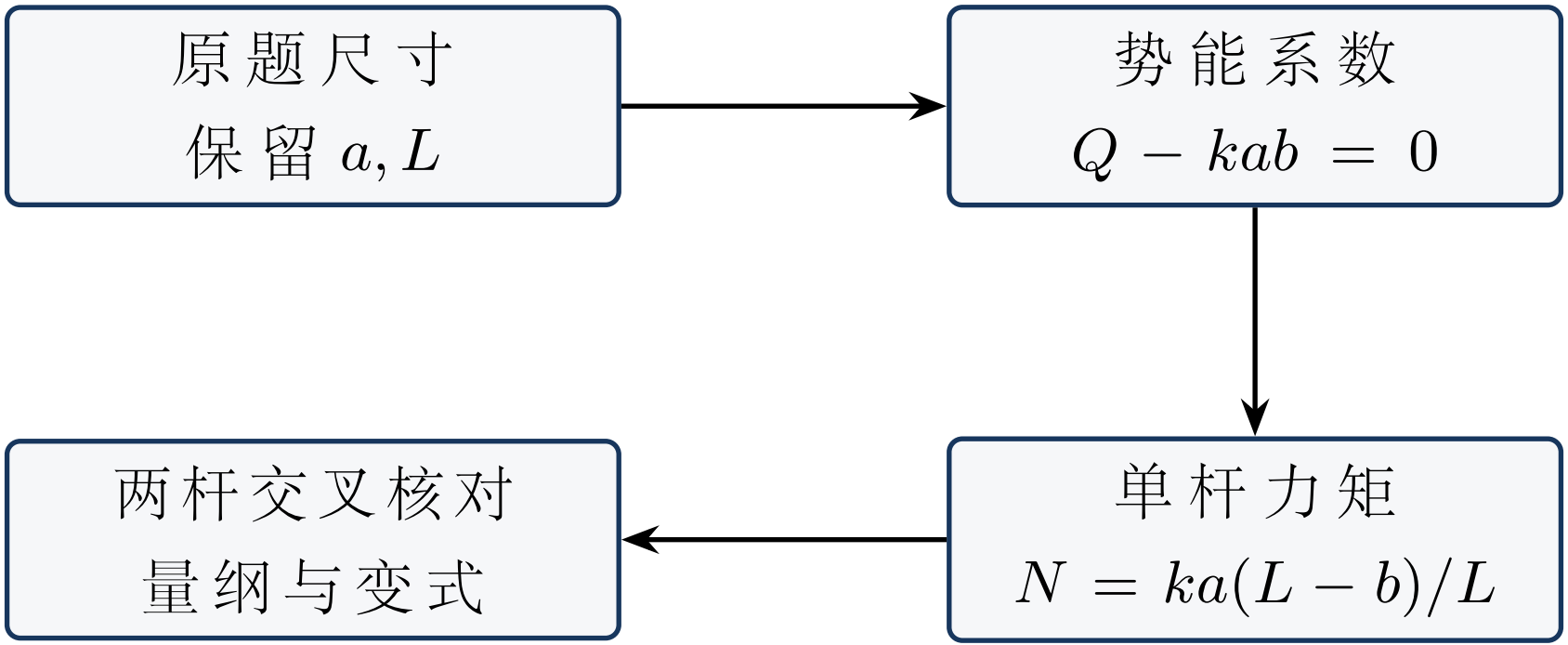
$$y_A = a + L\cos\theta, \quad y_{CD} = L\cos\theta/2$$



$$U = U_0 + [gL(P_1 + P_2/2) - kab]\cos\theta$$



AD 为受压二力杆；$N = ka(L - b)/L$



$k_0 = 400L/a$；不得把未给出的 AD 长度当作已知